3D Print Scaler

User guide · IngeTrazo plugin · `print_scaler.py` · Ctrl+Alt+P

Scales a selected object to a model scale such as 1:50 or 1:76.2, shows its printed size in millimetres before you commit, and checks it against your print bed.

Installing and updating

1. Open File Explorer, paste `C:\Users\eddy_\AppData\Roaming\ingetrazo\plugins` into the address bar and press Enter.
2. Copy `print_scaler.py` into that folder. When updating, choose **Replace the file**.
3. Restart IngeTrazo. Settings are remembered only while IngeTrazo is open, so a restart also clears old settings.

Keep only one copy of the plugin in the folder. Two copies under different names may both load.

Shortcut: Ctrl+Alt+P

From the Python console: paste the file's code in, select what the plugin needs, then type `run()`. If it can't find the 3D view by itself, use `run(viewport)`.

Using it

1. Select the object: a group, or all of its faces.
2. Press **Ctrl+Alt+P**, check the printed size and click OK.

Setting	Effect
Scale	Presets from 1:1 to 1:1000, or **Custom...** for any scale such as 1:76.2. The custom box shows the chosen scale's number, so it starts from there
Model is drawn in	Leave this on Metres: IngeTrazo stores geometry in metres even when it displays millimetres. Marked **(guessed)** when the plugin guessed from the object's size; your own choice is remembered
Scale about	Bottom centre (stays on the print bed), centre, lowest corner, or the world origin
Mode	**Scale a copy** keeps the original and puts the copy beside it. **Scale in place** changes the original
Check against print bed	Your bed's X, Y and Z size in mm; the size line turns red if the copy won't fit, even turned 90°

The information line shows the current size and the printed size, for example "Printed size: 20.0 × 20.0 × 20.0 mm".

Scale in place

Scale in place refuses loose faces that share corners with geometry you didn't select, because moving them would distort the neighbours; group the object first or use **Scale a copy**. If any point can't be moved, the model is put back exactly as it was.

Printing workflow

1. Build at full size and make every part a closed solid.
2. Scale with **Scale a copy**, and check the bed fit.
3. Export the copy as STL.
4. In the slicer, check the **sliced layer preview**, not just the 3D view.

Detail size: at small scales, fine details shrink below what a printer nozzle can draw. A 10 mm mortar joint is 0.2 mm at 1:50, half the width of a standard 0.4 mm nozzle. Exaggerate small details before scaling, print larger, or use a finer nozzle.

Troubleshooting

Problem	Cause	Fix
The copy is 1000 times too big or small	Wrong drawing unit	Set **Model is drawn in** to Metres
"Nothing Selected" warning	No group or faces selected	Select the object before pressing Ctrl+Alt+P
"Joined to other geometry" message	Loose faces share corners with other parts	Group the object first, or use **Scale a copy**
Details missing in the print	Smaller than the nozzle	Exaggerate them before scaling, or print larger